of physics ideas, and as his greatest teacher—though Zel'dovich had never taken a formal course from him.

It required only a few months for Sakharov and Ginzburg, in Tamm's team, to come up with a far better design for a superbomb than the "prescription for failure" that Zel'dovich and the Americans were pursuing. Sakharov proposed constructing the bomb as a *layered cake* of alternating shells of a heavy fission fuel (uranium) and light fusion fuel, and Ginzburg proposed for the fusion fuel lithium deuteride (LiD). In the bomb's intense blast, the LiD's lithium nuclei would fission into tritium and helium nuclei, and the tritium would then fuse with the LiD's deuterium to form helium nuclei, releasing enormous amounts of energy. The heavy uranium would strengthen the explosion by preventing its energy from leaking out too quickly, by helping compress the fusion fuel, and by adding fission energy to the fusion. When Sakharov presented these ideas, Zel'dovich grasped their promise immediately. Sakharov's layered cake and Ginzburg's LiD quickly became the focus of the Soviet superbomb effort.

To push the superbomb forward more rapidly, Sakharov, Tamm, Belen'ky, and Romanov were ordered transferred from Moscow to the Installation. But not Ginzburg. The reason seems obvious: Three years earlier, Ginzburg had married Nina Ivanovna, a vivacious, brilliant woman, who in the early 1940s had been thrown into prison on a trumped-up charge of plotting to kill Stalin. She and her fellow plotters supposedly were planning to shoot Stalin from a window in the room where she lived, as he passed by on Arbat Street below. When a troika of judges met to decide her fate, it was pointed out that her room did not have any windows at all looking out on Arbat Street, so in an unusual exhibition of mercy, her life was spared; she was merely sentenced to prison and then to exile, not death. Her imprisonment and exile presumably were enough to taint Ginzburg, the inventor of the LiD fuel for the bomb, and lock him out of the Installation. Ginzburg, preferring basic physics research over bomb design, was pleased, and the world of science reaped the rewards: While Zel'dovich, Sakharov, and Wheeler concentrated on bombs, Ginzburg solved the mystery of how cosmic rays propagate through our galaxy, and with Landau he used the laws of quantum mechanics to explain the origin of superconductivity.

In 1949, as the Soviet atomic bomb project reached fruition, Stalin ordered that the full resources of the Soviet state be switched over, without pause, to